

$PSL_2(q)$ on $\mathbf{P}^1(\mathbf{F}_q)$

Exact sigma–Molien formula, transitivity boundary, and verification

Computational verification note

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Outcome. Formula (3.3) in the supplied statement is correct. Prime-field groups $q = 2, 3, 5, 7$ were constructed from all determinant-one 2×2 matrices and then quotiented by their projective action. All cycle-type multiplicities, 48 coefficients through degree four, and direct orbit counts agree. Determinant-level error is at most 6.67×10^{-16} .

1 Representation and coefficient meaning

Let $q = p^f$, $d = \gcd(2, q - 1)$, and $G = PSL_2(q)$, so $|G| = q(q^2 - 1)/d$. Let $V = \mathbb{C}[\mathbf{P}^1(\mathbf{F}_q)]$, of dimension $q + 1$. For a permutation with c_m cycles of length m ,

$$\det(I - tg)^{-1} = \prod_m (1 - t^m)^{-c_m}, \quad \prod_j \det(I - t_j g)^{-1} = \text{Exp}_\sigma \left(\sum_m c_m p_m \right). \quad (1)$$

The coefficient at τ equals $\dim(\bigotimes_s \text{Sym}^{\tau_s} V)^G$ by the Reynolds-projector argument. Because V has a basis permuted by G , it also equals the number of G -orbits on tuples of multisets having sizes τ_s . This gives a genuinely independent combinatorial check.

2 Four element types

Put $L_- = (q - 1)/d$ and $L_+ = (q + 1)/d$.

type	cycle structure on $\mathbf{P}^1$	multiplicity
identity	1^{q+1}	1
nonidentity unipotent	$1^1 p^{q/p}$	$q^2 - 1$
split semisimple, order $m > 1$	$1^2 m^{(q-1)/m}$	$\frac{q(q+1)}{2} \varphi(m), m \mid L_-$
nonsplit semisimple, order $m > 1$	$m^{(q+1)/m}$	$\frac{q(q-1)}{2} \varphi(m), m \mid L_+$

The unipotent form $x \mapsto x + 1$ fixes infinity and partitions the affine line into q/p cycles. A split element has two eigenlines; a nonsplit element has none. Each nontrivial semisimple element determines its eigenline pair, hence its unique torus. The split and nonsplit torus counts are $q(q + 1)/2$ and $q(q - 1)/2$ respectively, giving the displayed multiplicities.

3 Exact formula

Substituting the four rows into (1) proves

$$\begin{aligned} \mathfrak{M}_{G,V} = \frac{1}{|G|} & \left[\text{Exp}_\sigma((q+1)p_1) + (q^2-1) \text{Exp}_\sigma\left(p_1 + \frac{q}{p}p_p\right) \right. \\ & + \frac{q(q+1)}{2} \sum_{\substack{m|L_- \\ m>1}} \varphi(m) \text{Exp}_\sigma\left(2p_1 + \frac{q-1}{m}p_m\right) \\ & \left. + \frac{q(q-1)}{2} \sum_{\substack{m|L_+ \\ m>1}} \varphi(m) \text{Exp}_\sigma\left(\frac{q+1}{m}p_m\right) \right]. \end{aligned} \quad (2)$$

The multiplicities sum to $|G|$, an immediate global consistency check.

4 Poisson comparison and the exact boundary

Two-transitivity forces the same orbit counts as the stable symmetric-group action through total degree two:

$$[m_{(1)}] = 1, \quad [m_{(2)}] = [m_{(1,1)}] = 2.$$

For ordered triples, equality patterns contribute $1 + 3$ orbits, while distinct triples contribute

$$\frac{(q+1)q(q-1)}{|G|} = d.$$

Thus

$$[m_{(1,1,1)}] = 4 + d = \begin{cases} 5 & q \text{ even,} \\ 6 & q \text{ odd.} \end{cases} \quad (3)$$

At degree four, equality patterns with 1, 2, 3, 4 distinct values give

$$[m_{(1,1,1,1)}] = 1 + 7 + 6d + d(q-2) = 8 + d(q+4). \quad (4)$$

The last term is the projective cross-ratio parameter. Consequently the raw sigma-MGF has no finite coefficientwise limit as $q \rightarrow \infty$.

5 Verification strategy

For each prime q tested, the implementation enumerates every $(a, b, c, d) \in \mathbf{F}_q^4$ with $ad - bc = 1$, induces the fractional-linear permutation on $\mathbf{F}_q \cup \{\infty\}$, and removes scalar-kernel duplicates. It compares the actual cycle-type histogram with the four-type theorem, extracts every coefficient through degree four, independently counts orbits on tuples of multisets, and finally compares determinant averages of the permutation matrices at $(t_1, t_2) = (0.07, 0.11)$. The construction is intentionally transparent and is presently implemented for prime fields. The proof and formula cover all prime powers; testing $f > 1$ would require a small finite-field layer, not a change to the group-theoretic argument.